💻 Software Engineering (SE) Roadmap (Week-wise Structured Plan)

🧠 PHASE 1: Programming & OOP Foundations

📅 Weeks 1–6

📘 Week 1: Introduction to Programming

- What is Software Engineering?
- SDLC (Software Development Life Cycle) overview
- Choose language:
  - Java / Python / C++

👉 Learn:

- Variables
- Data types

📘 Week 2: Control Flow

- if / else
- Logical operators

👉 Practice:

- Simple decision-based programs

📘 Week 3: Loops

- for loop
- while loop

👉 Practice:

- Patterns
- Basic logic problems

📓 Week 4: Functions

- Modular programming
- Code reuse

📓 Week 5: OOP Basics

- Classes & Objects
- Constructors

📓 Week 6: OOP Advanced

- Inheritance
- Polymorphism
- Encapsulation

👉 Mini Project:

- Library management system

🧠 PHASE 2: Data Structures + Problem Solving

📅 Weeks 7–14

📓 Week 7: Arrays & Strings

- Basic operations

📒 Week 8: Linked Lists

- Insert / delete

📒 Week 9: Stack & Queue

- Implementation
- Use cases

📒 Week 10: HashMaps

- Fast lookup

📒 Week 11: Trees

- Binary trees
- Traversals

📒 Week 12: Graphs

- BFS / DFS

📒 Week 13: Algorithms

- Sorting
- Searching

📒 Week 14: Problem Solving Practice

- Daily coding practice
- Focus on patterns

PHASE 3: Software Engineering Core Concepts

Weeks 15–20

Week 15: SDLC Models

- Waterfall
- Agile

Week 16: Requirements Engineering

- Functional vs Non-functional requirements

Week 17: Software Design

- UML diagrams
- System modeling

Week 18: Design Patterns

- Singleton
- Factory
- MVC

Week 19: Software Architecture

- Monolith vs Microservices

Week 20: Documentation

- SRS (Software Requirement Spec)
- Technical docs

⍰ PHASE 4: Development Skills (Full Stack Focus)

📅 Weeks 21–32

📘 Week 21–22: HTML & CSS

- Web structure
- Styling

👉 Project:

- Portfolio website

📘 Week 23–24: JavaScript

- Basics
- DOM manipulation

📘 Week 25–26: Advanced JS

- Async programming
- APIs

📘 Week 27–28: Frontend Framework

- React (recommended)

📘 Week 29–30: Backend Development

- Node.js / Django
- REST APIs

📓 Week 31–32: Database Integration

- SQL
- Connect backend + DB

👉 Project:

- Blog or CRUD app

🧪 PHASE 5: Software Testing & DevOps

📅 Weeks 33–40

📓 Week 33–34: Software Testing

- Unit testing
- Integration testing

📓 Week 35–36: Debugging

- Finding bugs
- Logging

📓 Week 37–38: Version Control

- Git & GitHub
- Branching strategies

📓 Week 39–40: CI/CD

- GitHub Actions
- Automated deployment

🟣 PHASE 6: Projects (Real-World Development)

📅 Weeks 41–48

📘 Week 41–42: Beginner Projects

- To-do app
- Calculator

📘 Week 43–44: Intermediate Projects

- Blog system
- Authentication system

📘 Week 45–46: Advanced Projects

- E-commerce system
- Dashboard app

📘 Week 47–48: Final Team Project

👉 Build:

- Full-stack app (team-based if possible)

🧠 PHASE 7: Job Preparation

📅 Weeks 49–52

📘 Week 49: Resume & Portfolio

- GitHub projects
- Clean resume

📓 Week 50: Interview Preparation

- DSA revision
- System design basics

📓 Week 51: Mock Interviews

- Practice coding + behavioral

📓 Week 52: Apply for Jobs

- Internships
- Junior developer roles

🕒 Weekly Routine

- 5 days learning
- 2 days building / coding

Daily:

- 3–4 hours recommended

🎯 Final Outcome (After 1 Year)

By Week 52:

✅ Strong programming + OOP
✅ Solid DSA foundation
✅ Full-stack development skills
✅ Testing + DevOps basics

✅ 4–6 real projects
✅ Job-ready for:

- Software Engineer
- Web Developer
- Backend Developer

🔥 Key Difference (SE vs CS)

- CS = Theory + Algorithms
- SE = Building real software + teamwork

👉 SE is more **practical + job-focused**